



Alcohol-Dependent Individuals Make Detrimental Decisions under Ambiguous and Perilous Conditions

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Abstract The present study assessed the decision-making abilities of patients with alcohol dependence (AD) under ambiguity or risk conditions. Forty-eight AD patients and 50 healthy controls (HC) were subjected to the Mini-Mental State Examination (MMSE), Iowa Gambling Task (IGT), and Game of Dice Task (GDT) systems for evaluating their decision-making ability. Age, duration of education, and MMSE scores were compared between the AD and HC groups. The communication in each block/choice and the group was evaluated along with the relationship between the trial block and choice. The basic characteristics were similar between both groups. Reduced mean net scores (IGT) and frequencies of choice were found for various alternatives (GDT) in AD patients as compared to the healthy controls. The net scores of first block (IGT test) and GDT test in AD patients were associated remarkably. These findings indicated a worse decision-making performance in AD patients compared to HCs under both ambiguity and risk conditions.

Keywords Alcohol · Dependence · Decision-making · Disadvantageous choices · Risk-taking

Alcohol is a common addictive substance. A prolonged consumption can lead to alcohol dependence (AD), which results in detrimental physical, psychological, and social consequences (Dickerson 2007; Nowakowska et al. 2007), and excessive usage can lead to liver

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disease, an altered immune system, nutritional deficiencies, and brain damage (Ende et al. 2013).

Multiple neuropsychological tests have demonstrated that alcoholics suffer from cognitive impairments. Moulton et al. (2005) demonstrated that both acute intoxication and post-intoxication with alcohol impairs memory. In addition, Tedstone & Coyle (2004) found that specific attention-demanding processes were impaired in sober alcoholics. Assessing adult rats selectively bred to prefer or dislike alcohol, Criado & Ehlers (2010) reported lower event-related oscillation energy in the parietal cortex in the former group compared to rats disliking alcohol. Furthermore, a substantial number of structural brain imaging studies correlated the long-term alcohol use with cerebral changes (Konrad et al. 2012; Mainz et al. 2012).

The ability to take decisions potentially encompasses outcome prediction, alternative assessment, activity selection, and gaining knowledge from responses and reactions. Equipped with an explicit and implicit knowledge of risk, the above ability can be further categorized based on its occurrence under risk or ambiguous conditions (Brand et al. 2006; Brand et al. 2007). The evaluation of the ability to take decisions under risk conditions evidently potentiates the reward and punishment probabilities. However, in the case of ambiguous conditions, the possibilities for reward and punishment are uncertain and may be learned through feedback. Furthermore, the ability in risk conditions depends on the lateral medial prefrontal cortex, whereas choices made under ambiguous conditions predominately rely on the orbitofrontal cortex (Huettel et al. 2006).

The Iowa Gambling Task (IGT) is a psychological tool measuring the ability to make decisions during ambiguity, whereas the Game of Dice Task (GDT) is utilized in explicit situations. The above two approaches are widely used for investigating psychoneurosis, for example, in polysubstance abuse (Goudriaan et al. 2005) and youths presenting an alcohol history in the family (Cservenka & Nagel 2012).

As data about risk taking in ambiguous situations are rare in the context of AD, herein, IGT was used to clarify the occurrence of shift effect from ambiguous to risk choice in AD individuals. In addition, GDT was used to determine the extent of the deficit in AD individuals to make decisions.

Methods

Patients

The present study was approved by the Anhui Medical University Ethics Committee, and written informed consent was obtained from all willing participants, without any financial compensation.

Forty-eight patients with clinically diagnosed AD, according to the Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (DSM-IV; American Psychiatric Association 1994), were recruited from a mental hospital in Hefei and the First Affiliated Hospital of Anhui Medical University. All the AD individuals were tested after a minimum abstinence of 2 weeks to ensure an alcohol-free status. Patients were consecutively recruited as per availability between January 2013 and October 2014. Fifty age-matched and education-matched healthy adults were recruited as controls (mainly relatives of the patients and staff of the First Affiliated Hospital of Anhui Medical University). All the subjects were right-handed and had more than 5 years of education. The average duration education of the control group was 10.7 ± 2.1 years (range 6–16 years); the AD group had 10.5 ± 1.8 years (range 6–16 years).

The inclusion criteria were the following: (i) overall cognitive function score > 24 as assessed with the Mini-Mental State Examination (MMSE) and (ii) stern dependence on the alcohol and craving score > 5 as estimated with the Michigan Alcohol Screening Test (MAST) in China. Exclusion criteria were the following: (i) current or previous psychological disorders such as anxiety and depression, (ii) present use of psychotropic medication, or (iii) obvious brain lesions, as determined by computed tomography (CT).

The tests were conducted by staff members using the double-blind approach. The participants completed all assessments within 1 week, and the same IGT and GDT were administered.

Measurements

MMSE is employed for evaluating the cognitive status in patients (Folstein et al. 1975). The test consists of a 30-item questionnaire scaling of eight functional aspects (visuospatial, executive function, naming, memory, attention, language, abstract, and orientation). In addition, two computer-assisted neuropsychological measurements (IGT and GDT) were carried out, using questionnaires and instructions previously translated into Mandarin Chinese.

IGT

The IGT, an automated computerized gambling method for data collection, assessed the ability to make decisions in ambiguous conditions (Chen et al. 2013; Xi et al. 2011). Each trial (decision) requires the participants to select one out of the four virtual card decks labeled 1, 2, 3, and 4. The card selected from decks 1 or 2 could result in an instant \$100 reward or maximally a potential \$1000 loss. The card from decks 3 or 4 could result in only \$50 gain or a maximum loss of \$75. Therefore, decks 1 and 2 are disadvantageous as they might lead to a potentially higher immediate gain, but a greater loss subsequently. On the other hand, although the immediate gains are often lower in decks 3 and 4 as compared to those of 1 and 2, the reduced loss amount results in a positive final balance.

The participants were non-acquainted with the guidelines governing profits and losses, but rather expected to infer from the reaction of each choice. Each subject began the task with \$2000 displayed on the screen. The total net score evaluating the task performance was obtained by subtracting the number of disadvantageous choices from the advantageous options. The 100 trials were equally divided into five blocks, and the net score of each assessed the altered decision-making ability during the task.

GDT

The ability to make decisions in risky conditions was assessed by a computerized GDT (Ma et al. 2013). Here, the participants received 18 chances for tossing a virtual dice to maximize the profits using a fabricated €1000 initial capital, by 1 of the 4 alternatives. Each option defined gain/loss as well as various winning probabilities: (1) €1000–1:6 for a single number, (2) €500–2:6 for a double number combination, (3) €200–3:6 for a triple number combination, and (4) €100–4:6 for a quadruple number combination. The first two choices with lower winning probability are considered risky decisions, whereas the latter two are categorized as

not risky. The net score was obtained by deducting the number of non-risky from risky choices. While the positive net scores showed a preference for non-risky options, the negative scores were predisposed towards risky options. We also assessed the technique for choosing one of the four different options (single, double, triple, or quadruple numbers). Additionally, the participant was cognized with the gain/loss, the altered amount of principal, and the left-over dice throws after each toss.

Statistical Analysis

SPSS 16.0 (SPSS, Chicago, IL, USA) was used for the statistical analyses. Independent samples *t* test assessed the differences in age, years of education, and MMSE scores between the AD and HC groups. The one-way repeated measure ANOVA was used with the trial blocks (in the IGT test) and choice (four different gain/loss alternatives in the GDT test) as within-subjects' factors and study group (AD or HC group) as between-subjects' factors, to examine the interaction of each block/choice and group. The LSD and Tamhane's tests were used to perform the post hoc comparisons. The non-parametric Spearman's correlations evaluated the relationship between the trial blocks of the participants (in the IGT test) and choice (in the GDT test). *P* value < 0.05 was considered statistically significant.

Results

Demographic Characteristics and Performance on the MMSE

Table 1 did not demonstrate any statistically significant differences between the AD and HC groups with respect to age (48.21 ± 5.89 vs. 47.70 ± 6.00 years; $P = 0.673$), years of education (10.46 ± 1.79 vs. 10.72 ± 2.06 , $P = 0.503$), or MMSE scores (28.10 ± 1.77 vs. 28.24 ± 1.80 , $P = 0.707$).

Behavioral Performance on the IGT

The learning curves for both groups were shown in Fig. 1. Clearly, the individuals with AD displayed reduced net scores compared to the healthy individuals, especially from block 3; the net scores in blocks 4 and 5 were significantly higher in HC compared to the individuals with AD ($P < 0.05$). The key effect of the block was statistically significant [$F(4, 384) = 10.09$; $P < 0.001$]. Similar results were obtained in the case of the primary effect of group [$F(1,$

Table 1 Demographic characteristics of the AD and HC groups

	AD	HC	<i>t</i>	<i>P</i>
Gender (M:F)	40/8	43/7	—	—
Age (years)	48.21 ± 5.89	47.70 ± 6.00	0.423	0.673
Education (years)	10.46 ± 1.79	10.72 ± 2.06	− 0.671	0.503
MMSE	28.10 ± 1.77	28.24 ± 1.80	− 0.377	0.707
MAST	12.69 ± 4.12	—	—	—

Data represent the mean $\pm$ standard deviation

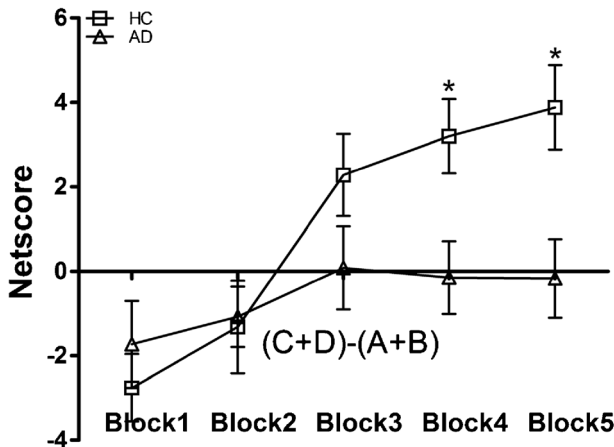


Fig. 1 IGT performance in the AD and HC groups. The mean net scores are presented for the five IGT blocks

96) = 4.07, $P = 0.047$]. In addition, the interaction between groups and blocks [$F(4, 384) = 3.70$, $P = 0.006$] showed significant differences (Table 2).

Behavioral Performance on the GDT

The AD group made disadvantageous choices more often than healthy controls. The frequencies of choice for each option in the GDT were determined and repeated measure ANOVA compared the “choice” (within-subject factor) and “group” (between-subject factor), revealing a remarkable interaction [$F(3, 288) = 15.29$, $P < 0.001$]. Moreover, the primary effect of choice [$F(3, 288) = 2.93$, $P = 0.034$] differed significantly. The AD patients selected single combination more frequently and triple combination more infrequently compared to healthy individuals: independent samples t test revealed considerable differences in the frequencies of choosing a single number ($t = 5.657$, $P < 0.001$), triple numbers ($t = -3.408$, $P = 0.001$), and quadruple numbers ($t = -2.577$, $P = 0.011$) between both groups. However, the frequency of

Table 2 Neuropsychological performances on the IGT and GDT with the number of selections

	AD	HC	t	P
IGT				
Block1	-1.73 ± 7.11	-2.76 ± 5.64	0.797	0.428
Block2	-1.08 ± 4.95	-1.32 ± 7.76	0.181	0.857
Block3	0.08 ± 6.80	2.28 ± 6.88	-1.589	0.115
Block4	-0.15 ± 5.98	3.20 ± 6.25	-2.707	0.008
Block5	-0.17 ± 6.47	3.88 ± 7.03	-2.963	0.004
Net score	-3.04 ± 18.79	5.28 ± 21.88	-2.016	0.047
GDT				
Single	7.25 ± 5.29	2.46 ± 2.76	5.657	< 0.001
Double	3.98 ± 3.30	4.56 ± 2.94	-0.920	0.360
Triple	4.08 ± 3.92	6.44 ± 2.87	-3.408	0.001
Quadruple	2.69 ± 3.71	4.54 ± 3.41	-2.577	0.011
Net score	-4.46 ± 10.15	3.96 ± 9.09	-4.330	< 0.001

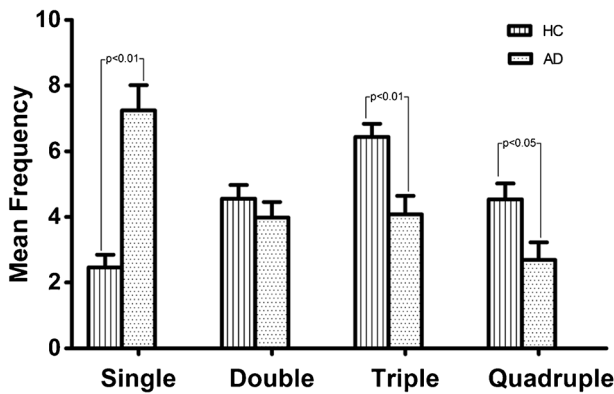


Fig. 2 The mean ($\pm$ SD) frequency of choices for the four different alternatives in GDT in the AD and HC groups

selecting double numbers ($t = -0.920$, $P = 0.360$) was similar between both groups (Fig. 2 and Table 2).

Correlation Analysis of Decision-Making Performance

The relationships of performances on blocks (IGT) and net scores (GDT) are summarized in Table 3. The correlation analysis within the AD group indicated significant associations between the net scores of the first and third ($r = 0.370$ and $P = 0.01$), second and third ($r = 0.580$ and $P < 0.001$), third and fourth ($r = 0.471$ and $P = 0.001$), fourth and fifth ($r = 0.322$ and $P = 0.026$), and the total net scores and first to fifth ($r = 0.506$, 0.591 , 0.885 , 0.596 , 0.359 ; all $P < 0.001$, except the last, $P = 0.012$) blocks of the IGT. The net scores of the first block of IGT and a total net score of GDT in the AD group were significantly associated.

Table 3 Correlations between the net scores of individual blocks on IGT and GDT

	Block1	Block2	Block3	Block4	Block5	IGT-net score
HC						
Block2	0.486**					
Block3	0.464**	0.487**				
Block4	0.257	0.187	0.432**			
Block5	0.155	0.016	0.371**	0.021		
IGT-total	0.727**	0.708**	0.804**	0.515**	0.422**	
Net score						
GDT-total	0.197	0.002	0.213	0.085	0.094	0.189
AD						
Block2	0.076					
Block3	0.370**	0.580**				
Block4	0.132	0.259	0.471**			
Block5	-0.207	0.119	0.169	0.322*		
IGT-total	0.506**	0.591**	0.885**	0.596**	0.359*	
Net score						
GDT-total	0.322*	0.026	0.067	-0.170	-0.278	0.042
Net score						

* $P < 0.05$

** $P < 0.01$

Healthy control subjects exhibited a striking association between the net scores of the first and second ($r = 0.486$ and $P < 0.001$), first and third ($r = 0.464$ and $P = 0.001$), second and third ($r = 0.487$ and $P < 0.001$), third and fourth ($r = 0.432$ and $P = 0.002$), third and fifth ($r = 0.371$ and $P = 0.008$), and total net scores and 1st–5th ($r = 0.727, 0.708, 0.804, 0.515, 0.422$, all $P < 0.01$) blocks of the IGT. Either of the groups did not reveal any relationship between the number of choices in the GDT and net scores of the IGT test; also, none was found between MMSE scores and decision-making performances.

Discussion

The current study demonstrated that AD patients were impaired with respect to the ability to make decisions under ambiguity and risk. In contrast to healthy controls, the AD group demonstrated disrupted decision-making ability by selecting disadvantageous choices, displayed by IGT and GDT.

Makris et al. (2008) revealed a significantly decreased reward-network volume in alcoholic subjects assessed by magnetic resonance imaging (MRI). Kahkonen et al. (2003) established the effect of alcohol as reduced excitability in the prefrontal cortex. Thus, several studies indicated that AD results in brain injury, especially in the prefrontal cortex, a tissue pivotal in the functions of execution and ability to make decisions (Ende et al. 2013; Konrad et al. 2012). Consequently, injured frontal cortex may impair the ability to take decisions potentially encompassing outcome prediction, alternative assessment, activity selection, and gaining knowledge from responses and reactions (Brand et al. 2006; Brand et al. 2007). In the case of ambiguous conditions, the possibilities for reward and punishment are uncertain and may be learned through feedback, which is probably impaired in AD (Brevers et al. 2014).

A significant correlation was established between the net scores obtained from the first block of IGT test and a total net score of GDT test in the AD group. In both AD and control groups, no significant correlations were observed amongst the selection numbers in GDT and net scores of IGT. These findings corroborate with previous studies (Dom et al. 2006; Noel et al. 2007). Furthermore, Brevers et al. (2014) showed that AD impairs the ability to decide optimally in multiple facets of uncertainty and that this deficit is associated with poor working memory processes. Cantrell et al. (2008) showed that poor decision making on the IGT was associated with dis-inhibitory disorders. These present study is also supported by results observed in other types of substance addiction (Brand et al. 2008).

Noel et al. (2007) found that AD participants demonstrated a shift in the decision-making mechanism of the IGT and were inclined to choose disadvantageous options; their study also suggested that AD patients encountered trouble using the former experience as a limited resource. These findings were confirmed by Rushworth & Behrens (2008) and Kim et al. (2011). In the current study, although the AD participants detected the shift earlier as compared to those previously studied, they continually showed learning deficits compared to the controls.

Goudriaan et al. (2005) suggested that AD participants' performance in the IGT resembles that of casino gamblers. Common personality traits between individuals with AD and casino gamblers, such as high sensation seeking, could explain this behavioral similarity. People with a high sensation-seeking personality often exhibit an abnormally high sensitivity to reward and stimuli that predict the rewards, leading to compulsive and persistent ingestion of alcohol. Such individuals may be attracted to a disadvantageous choice characterized by higher reward. In this study, the AD

participants made disadvantageous selections in the last two blocks of IGT, indicating reckless behaviors. In the GDT, AD participants chose more disadvantageous options and with higher frequencies. What's more, the frequencies of choosing single or double numbers were significantly higher in the AD participants. The IGT and GDT data suggested that the impairment in the ability to make decisions in AD participants could be ascribed to a danger-seeking personality.

In conclusion, the individuals with AD were detected functioning on a decision-making mechanism in the IGT and GDT tasks. In addition, the AD individuals selected more disadvantageous choices. These findings contribute to our knowledge of alcohol dependence and provide insight for improving therapeutic interventions.

Compliance with Ethical Standards

Ethical Approval This study was approved by the Anhui Medical University Ethics Committee, and written informed consent was obtained from all participants, who voluntarily participated in the trial, with no financial compensation awarded for their performance.

Informed Consent Informed consent was obtained from all individual participants included in the study.

Conflict of Interest The authors declare that they have no any conflict of interests.

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